

MULUND COLLEGE OF COMMERCE (AUTONOMOUS)

Programme: School of Business (Business Administration)

Class: FYBBA

Semester: I

Exam: External

Name of the Subject: Practical Approach to Mathematics

Date of Exam: 11/10/2025

Marks: 30

Time: 60 Minutes

Instructions: 1) Figures to the right indicate maximum marks.

Q1 Attempt either (a) and (b) OR (c) and (d) of the following.

- a) The accumulated amounts for a certain principal, with compound interest at a certain rate in 2 years and in 4 years are ₹ 12100 and ₹14641. Find the principal and rate of interest. 8
- b) What amount would be accumulated at the end of 2 years if ₹3600 is deposited at the end of each year at the rate of 11%p.a. 7

OR

- c) Find the present value of an immediate annuity of ₹4500 p.a. for 4 years at 8% p.a. 8
- d) What sum of Money will give a simple interest of ₹ 5200 at a rate of 5% p.a. after 2 years? 7

Q2 Attempt either (a) and (b) OR (c) and (d) of the following.

- a) If $A = \begin{bmatrix} 5 & 1 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -2 \\ -1 & 1 \end{bmatrix}$ show $(AB)^T = B^T A^T$. 8
- b) If $\begin{bmatrix} x+3 & z+4 & 2y-7 \\ -6 & a-1 & 0 \\ b-3 & -21 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 6 & 1 \\ -6 & -3 & 2c+2 \\ 5 & -21 & 0 \end{bmatrix}$,
find the values of a, b, c, x, y, z 7

OR

- c) If $A = \begin{bmatrix} 1 & 3 & 0 \\ 4 & -1 & 9 \\ 2 & 0 & 3 \end{bmatrix}$ find $A^2 - A + I$ 8
- d) i) Solve the following system of equations by Cramer's Rule (04)
 $5x + 3y = 17; 3x + 7y = 31$
ii) Find the value of $\begin{vmatrix} 3 & 6 & 11 \\ 2 & 7 & 11 \\ 2 & 6 & 12 \end{vmatrix}$ (03) 7

